

Multiple Choice (10 marks)

1. You've made a scientific theory that there is an attractive force between all objects. When will your theory be proven to be correct?
 - (a) When the theory can predict future observations.
 - (b) As soon as you propose the theory.
 - (c) When your theory becomes a law.
 - (d) You can never prove your theory to be correct only "yet to be proven wrong".
 - (e) When the scientific community accepts it as a fact.
2. According to Coulomb, a pair of charged particles placed twice as close to each other experience a force
 - (a) four times as strong
 - (b) half as strong
 - (c) one-fourth as strong
 - (d) inversely proportional to the distance
 - (e) twice as strong
3. A particle of charge $1.8\mu\text{C}$ is at the center of a Gaussian cube 55 cm on edge. What is the net electric flux through the surface?
 - (a) $2.2 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$
 - (b) $2.8 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$
 - (c) $3.2 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$
 - (d) $1.0 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$
 - (e) $1.510^5 \text{ N} \cdot \text{m}^2/\text{C}$
4. Which statement about an atom is not true:
 - (a) A neutral atom always has equal numbers of electrons and protons.
 - (b) The electrons can only orbit at particular energy levels.
 - (c) Electrons and protons in a neutral atom are always equal in number.
 - (d) Electrons can orbit at any energy level.
 - (e) A neutral atom always has equal numbers of neutrons and protons.
5. The plates of a capacitor are charged to a potential difference of 5 V. If the capacitance is 2 mF, what is the charge on the positive plate?

- (a) 2 C
- (b) 0.02 C
- (c) 0.02 mC
- (d) 0.005 C
- (e) 0.1 C

True / False (10 marks)

Answer True or False

6. True or False: The cost to run a motor that draws 5 amperes from a 120-volt line for 2 hours is 15 cents.
7. True or False: For a particle moving with constant velocity along a cardioid, the radial component of acceleration is given by $-(3/4)(v^2/k)$.
8. True or False: The Mars Exploration Rover Spirit is tilted towards the north because it is in the southern hemisphere where it is currently winter.
9. True or False: The electrostatic force between two positive charges of 100C and 200C separated by 3 cm is 5×10^5 N.
10. True or False: The electric field produced by a finite-length charged rod decreases as the inverse square of distance at large distances from the rod.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Given point charges of $+6.0 \mu\text{C}$ at $x=8.0$ m and $-4.0 \mu\text{C}$ at $x=16.0$ m, determine the required charge at $x=24.0$ m for a charge at the origin to experience no net electrostatic force. Explain your reasoning and calculations.
12. Given a 1.4447-gm sample of coal burned in an oxygen-bomb calorimeter, analyze the temperature rise and the system's water equivalent to calculate the heating value of the coal. Discuss the implications of your findings.

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